

Machine Learning-Driven Cross-Sell Prediction for Insurance Customers

First Review Report

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1.ABSTRACT

This project focuses on predicting whether existing health insurance customers are likely to purchase vehicle insurance, thereby enabling effective cross-selling. Using a dataset of 381,109 customer records from Kaggle, the study applies machine learning classification techniques such as Logistic Regression, Decision Trees, XGBoost, CatBoost, and Gradient Boosting. Data preprocessing, feature engineering, and class balancing methods are used to enhance model accuracy.

The outcome of this study supports insurance companies in improving marketing strategies, customer engagement, and overall revenue by providing data-driven insights for personalized cross-sell recommendations. Recent studies (Mavundla et al., 2024; Galvão & Moral, 2025) confirm that machine learning approaches remain highly effective for cross-sell prediction, which justifies the continued use of this problem and dataset.

2.EXISTING SYSTEM

Insurance companies have traditionally relied on conventional methods such as generic telemarketing and bulk email campaigns to promote cross-sell opportunities. While cost-effective at scale, these approaches lack personalization and typically lead to poor conversion rates [1].

Some insurers continue to use basic statistical models, such as frequency counts and demographic grouping, for customer segmentation. Although simple to implement, these methods fail to capture complex behavioral patterns, resulting in low predictive accuracy [2].

Manual risk assessment and expert-driven segmentation are still practiced in certain cases. However, these approaches are prone to bias, inconsistency, and scalability challenges, limiting their effectiveness in today's data-driven insurance environment [3].

Limitations of Existing Systems:

Low accuracy – because conventional statistical/telemarketing models can't capture complex customer behavior.

High marketing cost – since bulk campaigns are wasteful and not well-targeted.

Lack of personalization – no customer-specific offers, only generic ones.

Customer dissatisfaction – customers ignore or get frustrated with irrelevant offers.

3.PROPOSED SYSTEM :

The proposed system uses Machine Learning to predict cross-sell potential. By analyzing customer demographics, past insurance history, vehicle details, and premium information, the system identifies customers most likely to purchase vehicle insurance.

FEATURES:

Predictive modeling using classification algorithms.
Automated feature selection and hyperparameter tuning.
Balanced learning with techniques like SMOTE.
Evaluation using accuracy, precision, recall, and AUC-ROC.

4.SYSTEM ARCHITECTURE :

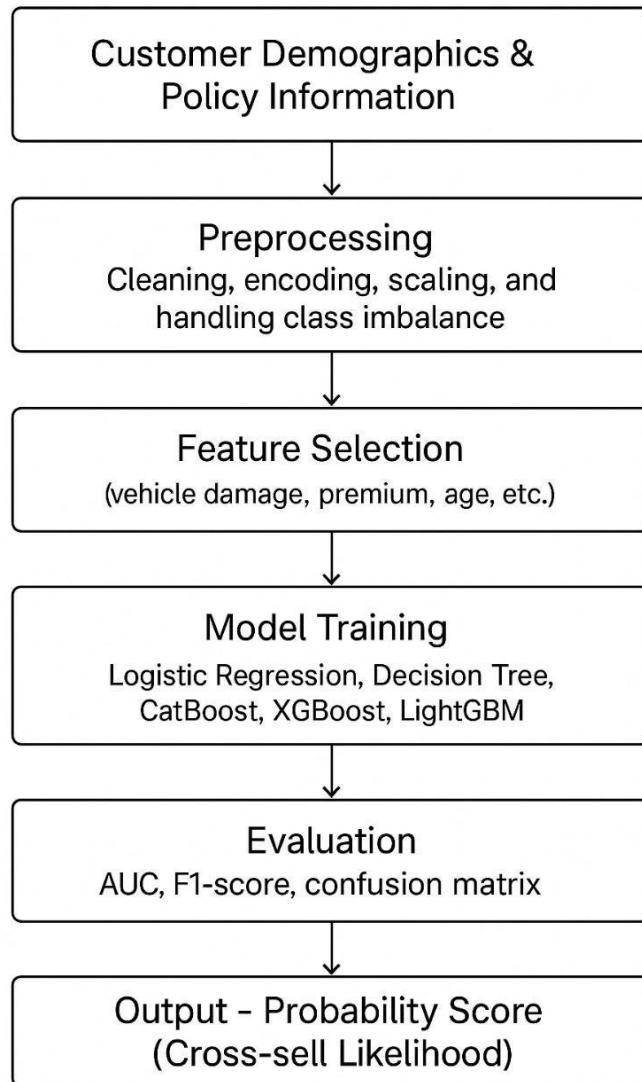


FIG 1.SYSTEM ARCHITECTURE FOR CROSS-SELL PREDICTION The system begins with input data containing customer demographics and policy details such as age, premium, and vehicle damage history. This data then undergoes preprocessing, which includes cleaning, encoding categorical variables, scaling numerical features, and handling class imbalance to make the dataset consistent and balanced. After preprocessing, feature selection is carried out to identify

the most important predictors, with key factors like age, premium amount, and vehicle damage chosen to improve accuracy. The refined dataset is then used in the model training phase, where multiple algorithms such as Logistic Regression, Decision Tree, CatBoost, XGBoost, and LightGBM are trained and fine-tuned. These models are evaluated using performance metrics like AUC, F1-score, and confusion matrix to compare their effectiveness. Based on this evaluation, the best-performing model is selected. Finally, the system outputs a probability score that indicates the likelihood of a customer accepting a cross-sell offer.

5.HARDWARE & SOFTWARE SPECIFICATION

Hardware Specifications:

- * Processor: Minimum i5 / Ryzen 5
- * RAM: 8 GB
- * Storage: 500 GB HDD / 256 GB SSD

Software Specifications

- * Python 3.x
- * Libraries: Pandas, NumPy, Scikit-learn, CatBoost, XGBoost, LightGBM, Matplotlib, Seaborn
- * IDE: Jupyter Notebook / PyCharm
- * Dataset Source: Kaggle

6. WORK PLAN

- 1.Data Collection – Extract dataset from Kaggle.
2. Data Preprocessing – Handle missing values, encode categorical data, scale features.
3. Exploratory Data Analysis (EDA) – Visualize age, premium distribution, and correlation.
4. Model Building– Train ML models (Logistic Regression, Decision Tree, Boosting methods).
5. Model Evaluation – Compare results using AUC-ROC and precision-recall.
6. Deployment Plan – API / Web integration for insurance companies.

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7. WORK DONE SO FAR

Week 1– Requirement gathering, problem statement finalized.

Week 2–3 – Data collection from dataset sources, preprocessing (cleaning, handling missing values, encoding).

Week 4 – Performed exploratory data analysis (statistical summary, distributions, outlier detection).

Week 5 – Visualized correlations, checked class imbalance, and identified key features for modeling.

REFERENCES:

[1] Mavundla, K., Thakur, S., Adetiba, E., & Abayomi, T. (2024). *Predicting Cross-Selling Health Insurance Products Using Machine-Learning Techniques*. *Journal of Computer Information Systems*.

[2] Chu, J. Z. K., Than, J. C. M., & Jo, H. S. (2024). *Deep Learning for Cross-Selling Health Insurance Classification*. *Proceedings of GECOST 2024*.

[3] Galvão, L. R., & de Andrade Moral, R. (2025). *Non-Heuristic Selection via Hybrid Regularized and Machine Learning Models for Insurance*. *Expert Systems with Applications*, 244, 122918.

Signature of the Student

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